

Mesoscopic Shift–Spectral Gaps and Tensor–Normalization Barriers for Three–Coefficient Determinant Packets

Liang Wang

School of Artificial Intelligence and Automation,
Huazhong University of Science and Technology, Wuhan 430074, P.R. China
wangliang.f@gmail.com

July 2026

Abstract

The large-prime reduction in the twin-prime-correlation (TPC) sequence admits scalar projective layers of the form $pr - mj = h$, with separated arithmetic weights on the prime p , a Möbius cofactor r , and an actual source variable m , while the orbit variable j is smooth up to a fixed residue-class selector. The physical Hilbert dual later recouples these variables. We separate three questions that are lost if this is described only as a “third-coefficient problem.” First, multiplicative synthesis and Cauchy–Schwarz close the scalar fixed-shift form, and a Fourier–Bohr lift also closes the residual label $s = d(m)r$, at the Cartesian tensor-energy scale. That scale is the sum of the atomic diagonals over all shifts, not the physical diagonal at one shift; flat boxes and an explicit fixed-shift construction show that the two normalizations may differ by a polynomial factor. Second, Poisson summation in the smooth orbit slot gives the exact kernel $K_m(\alpha) = J \sum_k \widehat{w}(J(k - m\alpha))$. Band-limited orbit weights therefore produce a genuine shift-spectral gap between the geometric envelope and the first sampling aliases. At the TPC endpoint this gap is $X^{-1+o(1)} < \|\alpha\| < X^{-267/400+o(1)}$; corresponding band-pass averages vanish, and low-pass averages have an exact main term. Restriction to any fixed orbit progression preserves this exponent gap. Third, weighted shift averaging leaves the exact fiber-grouping norm unchanged when numerator and diagonal use the same weights. Moreover, sparse TPC alias sampling folds the full shift spectrum, and the currently proved collision-free local-label range does not reach the Poisson scale. Thus the paper proves a deterministic spectral advance and identifies atomic normalization, orbit-output coupling, alias folding, and fixed-shift localization as separate remaining gates. No fixed-shift TPC closure, parity breach, or twin-prime conclusion is claimed.

Keywords: determinant equation; shift spectrum; Möbius correlation; Poisson summation; Fourier–Bohr transform; tensor normalization; parity problem.

Convention. We write $e(x) = \exp(2\pi ix)$, identify $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, and put $\|\alpha\|_{\mathbb{T}} = \min_{k \in \mathbb{Z}} |\alpha - k|$. All arithmetic supports at one scale are finite. Fourier transforms on $\mathbb{R}$ use $\widehat{f}(\xi) = \int_{\mathbb{R}} f(x)e(-x\xi) dx$.

MSC 2020: Primary 11L07, 11N35; Secondary 42A38, 46B28, 11N36.

1 Introduction

The TPC sequence isolates a deterministic equality-column problem whose large-prime part is governed by

$$pr - mj = h_0. \tag{1}$$

Here p is prime, r is the complementary factor carrying a Möbius weight, m is an actual source row carrying a second arithmetic coefficient, and j is an orbit variable with a smooth profile

restricted to a fixed residue class. TPC-45 identifies the absence of a third arbitrary determinant-coefficient slot in the available fixed-determinant estimates as the next analytic obstruction [7]. The natural response is to average the determinant shift and use the progression-smooth j -slot. The purpose of this paper is to determine exactly what that operation proves, which normalization it controls, and why it does not yet return the physical column at h_0 .

There are three families that must not be merged. A *frozen completed family* keeps one coefficient tensor and lets the determinant value range through consecutive integers. The inherited TPC alias bank keeps one coefficient tensor but samples that family on the progression

$$h_0 + kM_{\text{al}}, \quad M_{\text{al}} = X^{399/400+o(1)}.$$

A family reconstructed at different physical shifts changes the eligible rows, residue classes, coprimality conditions, and generally the coefficient tensor itself. Fourier analysis applies directly to the first family. It applies to the second only after alias folding, and it does not become a theorem about the third without a uniform provenance statement.

1.1 Main results

Our first result is an exact positive closure theorem at a deliberately named norm. For scalar separated coefficients, put

$$\mathcal{T}_h = \sum_{pr-mj=h} a_p b_r c_m \eta_j, \quad \mathcal{D}_{\otimes} = \left(\sum_{p,r} |a_p b_r|^2 \right) \left(\sum_{m,j} |c_m \eta_j|^2 \right).$$

Multiplicative synthesis turns $\mathcal{T}_h$ into an additive correlation. The resulting Cauchy–Schwarz estimate is

$$|\mathcal{T}_h|^2 \leq \rho_{pr} \rho_{mj} \mathcal{D}_{\otimes}. \quad (2)$$

At the large-prime TPC interface, $\rho_{pr} = 1$ and $\rho_{mj} = X^{o(1)}$. A finite Fourier–Bohr transform gives the same tensor-scale closure after retaining the residual label $s = d(m)r$. These are genuine fixed-shift theorems, but

$$\mathcal{D}_{\otimes} = \sum_h \Delta_h, \quad (3)$$

where Δ_h is the atomic diagonal at shift h . The physical problem is normalized by Δ_{h_0} , not its sum over all shifts. Flat boxes and the prescribed-shift construction in [section 8](#) show that the two scales can differ by a power of X . Hence (2) is not renamed as an atomic-normalized TPC estimate.

Our second result is a deterministic shift-spectrum theorem. For a Hilbert-valued frozen packet with a j -independent vector coefficient, the shift Fourier transform factors through

$$K_m(\alpha) = \sum_{j \in \mathbb{Z}} w(j/J) e(mj\alpha) = J \sum_{k \in \mathbb{Z}} \widehat{w}(J(k - m\alpha)). \quad (4)$$

If $\text{supp } \widehat{w} \subset [-\sigma, \sigma]$, this kernel vanishes in the exact annulus

$$\frac{\sigma}{M_- J} < \|\alpha\|_{\mathbb{T}} < \frac{1 - \sigma/J}{M_+}. \quad (5)$$

At the endpoint this is

$$X^{-1+o(1)} < \|\alpha\|_{\mathbb{T}} < X^{-267/400+o(1)}.$$

Restriction to a fixed progression in j changes only the constant at the first alias and preserves these endpoint exponents. Compactly supported smooth orbit profiles satisfy rapid decay on every fixed power-shrinking of the same annulus. As consequences, suitably supported band-pass

shift averages vanish exactly, while sufficiently coarse low-pass averages admit an exact zero-mode formula. The smooth version has a power-saving Poisson remainder. We also compute the corresponding averaged atomic diagonal and keep it distinct from the full squared synthesis.

Our third result is a set of sharp transfer boundaries. Weighted averaging in h leaves the exact fiber-grouping norm equal to the largest realized fiber when numerator and atomic diagonal use the same weights. Sampling only the TPC alias progression folds the entire shift spectrum into M_{al} branches. For actual reconstructed shifts, zero extension always gives a common ambient coordinate space, while the present rigidity argument proves collision-free equality labels throughout

$$|h - h_0| \leq X^{29629/210000-\epsilon}, \quad (6)$$

and makes no assertion that this sufficient range is optimal. Resolving the first Poisson alias by the present route needs a dense window of length at least $X^{267/400+o(1)}$. Finally, positivity alone costs the full effective number of shifts when an averaged nonnegative quantity is localized back to one chosen shift. These statements separate four missing inputs: orbit-frequency regularity for the pulled-back dual field, atomic rather than Cartesian normalization, anti-alias control, and fixed-shift localization.

1.2 Why existing trilinear estimates do not fill the gap

Bettin–Chandee’s trilinear Kloosterman estimate permits three arbitrary sequences, but one is a Kloosterman-numerator sequence [1]. Their fixed-determinant corollary retains two smooth determinant slots and only two arbitrary arithmetic slots. Averaging the determinant in the Bettin–Chandee–Radziwiłł application compresses a frequency and a determinant into a structured numerator sequence after smooth Poisson analysis; it does not remove those two smooth-slot hypotheses [2]. The modulus-dispersion results of Fouvry–Radziwiłł and Wright concern different averaged objects [3, 8]. Moreover, the physical TPC dualization introduces the joint field

$$z_{\text{output row}, j, d(m)r},$$

which need not split into their scalar coefficient slots. The precise comparison is given in [section 10](#).

1.3 Scope and organization

All structural identities, finite operator norms, Fourier support statements, Poisson formulas, and rational exponent comparisons stated as theorems are proved without a probabilistic Möbius hypothesis. Statements about the actual TPC packet use only the inherited coefficient interfaces explicitly cited from TPC–36, TPC–42, TPC–43, and TPC–45 [4–7]. No theorem asserts that the arbitrary dual field is orbit-band-limited, that the sparse alias bank can be unfolded at no cost, or that a shift average controls the physical column.

[Section 2](#) reconstructs the exact TPC packet and its dual coupling. [Sections 3 to 5](#) prove the scalar and residual tensor closures. [Sections 6 and 7](#) establish the spectral-gap and averaging formulas. [Sections 8 and 9](#) prove the sharp normalization, folding, geometry, and localization boundaries. [Section 10](#) audits the available external estimates, and [section 11](#) states four reversible next doors. The exact finite certificate and final claim ledger appear in [section 12](#).

No result in this paper proves a fixed-shift TPC closure, a Hardy–Littlewood prime-pair asymptotic, a breach of the sieve parity barrier, or the twin-prime conjecture.

2 The exact TPC determinant interface

This section records the coefficient packet to which the later scalar and tensor arguments are applied. There are three distinct objects in play: the physical equality column at one fixed shift,

the completed alias family inherited from the determinant completion, and a family of TPC packets reconstructed at genuinely different physical shifts. The first two share one source–mask coefficient field. The third generally does not. Keeping this distinction explicit prevents a statement about an auxiliary completion from being read as an averaged prime-pair statement.

2.1 Rows, outputs, and the physical terminal column

Fix a nonzero physical shift $h_0 \in \mathbb{Z}$. Let $\mathcal{A}_1$ and $\mathcal{A}_2$ be the two inherited source-row systems. A row $\rho \in \mathcal{A}_s$ has the form

$$\rho = (\ell_\rho, d_\rho), \quad m_\rho = \ell_\rho d_\rho \asymp Q, \quad \ell_\rho \asymp L_0, \quad d_\rho \asymp D_0, \quad (7)$$

where ℓ_ρ is prime, d_ρ is squarefree, and the integer m_ρ determines ρ . The orbit set $\mathcal{J}$ consists of $O(J)$ positive integers with $j \asymp J$. At the endpoint,

$$Q = X^{267/400+o(1)}, \quad J = X^{133/400+o(1)}, \quad QJ \asymp X, \quad (8)$$

$$D_0 = X^{10049/52500+o(1)}, \quad L_0 = X^{99979/210000+o(1)}. \quad (9)$$

The two output sets and their common Hilbert space are

$$\mathcal{I}_L = \{(L, \gamma, j) : \gamma \in \mathcal{A}_2, j \in \mathcal{J}\}, \quad \mathcal{I}_R = \{(R, \alpha, j) : \alpha \in \mathcal{A}_1, j \in \mathcal{J}\}, \quad \mathcal{H}_{\text{out}} = \ell^2(\mathcal{I}_L \sqcup \mathcal{I}_R). \quad (10)$$

Write the physical source coefficients as

$$\gamma_\rho^{(s)} = \mu(d_\rho) \gamma_\rho^{\circ, (s)}, \quad s = 1, 2. \quad (11)$$

Thus $\gamma^{\circ, (s)}$ contains the source logarithm, smooth cutoffs, and fixed residue data, but not the source Möbius sign. Let $\mathfrak{A}_{\alpha, \gamma}^{\text{act}}(j)$ be the actual joint source–mask multiplier and put

$$\Lambda(n) := (\log n) \Phi_0(n/X). \quad (12)$$

As usual, every terminal expression is set equal to zero outside the declared positive support. The physical left and right equality columns are

$$[\mathcal{J}_{h_0}^L]_{\gamma, j} := - \sum_{\alpha \in \mathcal{A}_1} \gamma_\alpha^{(1)} \mathfrak{A}_{\alpha, \gamma}^{\text{act}}(j) \mu(n_{\alpha, j}) \Lambda(n_{\alpha, j}), \quad (13)$$

$$[\mathcal{J}_{h_0}^R]_{\alpha, j} := - \sum_{\gamma \in \mathcal{A}_2} \gamma_\gamma^{(2)} \mathfrak{A}_{\alpha, \gamma}^{\text{act}}(j) \mu(n_{\gamma, j}) \Lambda(n_{\gamma, j}), \quad (14)$$

where

$$n_{\rho, j} = m_\rho j + h_0. \quad (15)$$

These formulas are the all-minus specialization of the TPC–43 prime sign model and agree with the stripped coefficients used in TPC–45 [6, 7].

Work on one nonempty dyadic terminal block

$$N_- \leq n_{\rho, j} \leq N_+, \quad N_- \asymp N_+ \asymp X. \quad (16)$$

For $R_0 < N_-/\sqrt{N_+}$, define a common large-prime band

$$\mathcal{P}_{R_0}^\# := \left\{ p \in \mathbb{P} : \sqrt{N_+} < p \leq \frac{N_-}{R_0} \right\}. \quad (17)$$

The subscript R_0 denotes a cofactor threshold and is unrelated to the right output symbol in $\mathcal{I}_R$.

Proposition 2.1 (Exact physical large-prime determinant sector). *Let $\mathcal{T}_{h_0, \#}^{L/R}$ denote the part of $\mathcal{T}_{h_0}^{L/R}$ for which the terminal target is divisible by a prime in $\mathcal{P}_{R_0}^\#$. Then*

$$[\mathcal{T}_{h_0, \#}^L]_{\gamma, j} = \sum_{\substack{p \in \mathcal{P}_{R_0}^\#, r, \\ \rho = (\ell, d) \in \mathcal{A}_1 \\ pr - \ell dj = h_0}} \gamma_\rho^{(1)} \mathfrak{A}_{\rho, \gamma}^{\text{act}}(j) \mu(r) \Lambda(pr), \quad (18)$$

$$[\mathcal{T}_{h_0, \#}^R]_{\alpha, j} = \sum_{\substack{p \in \mathcal{P}_{R_0}^\#, r, \\ \rho = (\ell, d) \in \mathcal{A}_2 \\ pr - \ell dj = h_0}} \gamma_\rho^{(2)} \mathfrak{A}_{\alpha, \rho}^{\text{act}}(j) \mu(r) \Lambda(pr). \quad (19)$$

Every terminal target occurs at most once in either sum, and every occurring cofactor satisfies

$$R_0 \leq r < \sqrt{N_+}. \quad (20)$$

In particular, the three nonsmooth arithmetic roles in the collapsed equation

$$pr - mj = h_0 \quad (21)$$

are the prime variable p , the cofactor weight $\mu(r)$, and the actual source–mask coefficient on the row variable m . The orbit variable j is smooth apart from the fixed residue selector recorded in [proposition 2.2](#).

Proof. If $p > \sqrt{N_+}$ divides a target $n \leq N_+$, it is the unique prime from $\mathcal{P}_{R_0}^\#$ dividing that target. Write $n = pr$. The upper endpoint in (17) gives $r \geq R_0$, and $p > \sqrt{N_+}$ gives $r < \sqrt{N_+}$. Also $p > r$, so $p \nmid r$. Consequently

$$-\mu(pr) = \mu(r), \quad (22)$$

including the case in which either side vanishes. Substitution into [equations \(13\) and \(14\)](#) proves the two formulas. Finally $m = \ell d$ determines its source row, so the pair (ℓ, d) may be collapsed to one sequence on m without losing any coefficient data. $\square$

The sign in (22) is worth fixing: the source coefficient $\gamma^{(s)}$ already contains $\mu(d)$. Thus the determinant sector contains the two physical Möbius roles $\mu(d)$ and $\mu(r)$, while the large prime contributes the minus sign that cancels the minus in the terminal atom.

2.2 The inherited projective separation

The actual multiplier has the fixed-shift structure

$$\mathfrak{A}_{\alpha, \gamma}^{\text{act}}(j) = \mathfrak{m}_{\text{act}}(\alpha, \gamma) \Xi_{\alpha, \gamma}(j) \mathcal{W}_{\alpha, \gamma}(j/J), \quad (23)$$

where, for $\alpha = (\ell_\alpha, d_\alpha)$ and $\gamma = (\ell_\gamma, d_\gamma)$,

$$\mathfrak{m}_{\text{act}}(\alpha, \gamma) = \mathbf{1}_{\ell_\alpha \neq \ell_\gamma} \mathbf{1}_{|m_\alpha - m_\gamma| > \Delta_X} \mathbf{1}_{(d_\alpha, d_\gamma) \leq G_X}. \quad (24)$$

Here Δ_X and G_X are the inherited row-separation and gcd thresholds from TPC–36. We write $\mathfrak{p}_\infty$ for its projective tensor norm and

$$\omega_* := \max\{\omega(d) : d \text{ occurs in a physical row}\}.$$

Only the following displayed $X^{o(1)}$ estimate for these inherited objects is used below. TPC–36 proves that the static mask has bounded projective norm

$$\mathfrak{p}_\infty(\mathfrak{m}_{\text{act}}) \ll \log(2Q)(3 + 2\sqrt{2})^{\omega_*} = X^{o(1)}, \quad (25)$$

and that the fixed residue and smooth factors may be included without a power loss [4]. We use the following direct consequence.

Proposition 2.2 (Actual row–output projective separation). *Let $H_0 = |h_0|$. For every fixed $B \geq 0$, the left coefficient has an exact representation*

$$\begin{aligned} \gamma_\alpha^{(1)} \mathfrak{A}_{\alpha, \gamma}^{\text{act}}(j) &= \sum_{a \bmod H_0} \mathbf{1}_{j \equiv a \pmod{H_0}} \int_{\Omega_a} c_{\omega, a}(m_\alpha) q_{\omega, a}(\gamma) \phi_{\omega, a}(j/J) d\nu_a(\omega), \end{aligned} \quad (26)$$

where $c_{\omega, a}$ is supported on the physical row integers and

$$\sum_{a \bmod H_0} \int_{\Omega_a} \|c_{\omega, a}\|_\infty \|q_{\omega, a}\|_\infty \|\phi_{\omega, a}\|_{C^B} d|\nu_a|(\omega) \ll_{B, h_0} X^{o(1)}. \quad (27)$$

The right coefficient has the symmetric representation. If the source integer must instead be opened as $m = \ell d$, then for every fixed opposite row one has an exact expansion

$$\gamma_{(\ell, d)}^{(1)} \mathfrak{A}_{(\ell, d), \gamma}^{\text{act}}(j) = \sum_{a \bmod H_0} \mathbf{1}_{j \equiv a \pmod{H_0}} \int \sum_{\varrho} b_\varrho(\ell) f_\varrho(d) \eta_\varrho(j/J) d\nu, \quad (28)$$

whose corresponding weighted signed mass is $X^{o(1)}$. The static Ferrers part of this latter cost is $O(1 + \log(2D_0))$, although the formal number of layers can be $O(D_0)$.

Proof. TPC–36 gives a global separation of $\mathfrak{A}^{\text{act}}$ into a source-row factor, an opposite-row factor, and a smooth orbit factor with the cost in (25). Multiply the source-row factor by $\gamma_\alpha^{(1)}$. Its pointwise size is $X^{o(1)}$, because the only growing elementary factor is a source logarithm. Source separation permits the resulting function of α to be written as a function of m_α . This proves equations (26) and (27). The last statement is the fixed-column Ferrers separation of TPC–36, including its physical row and orbit weights. $\square$

Because Φ_0 is a fixed smooth terminal cutoff, Mellin inversion also separates the target product. We use the convention

$$\tilde{\Phi}_0(t) := \int_0^\infty \Phi_0(x) x^{-it} \frac{dx}{x}, \quad \Phi_0(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \tilde{\Phi}_0(t) x^{it} dt. \quad (29)$$

If

$$d\sigma(t) = \frac{1}{2\pi} \tilde{\Phi}_0(t) X^{-it} dt, \quad (30)$$

then

$$\Lambda(pr) = \int_{\mathbb{R}} ((\log p) p^{it} r^{it} + p^{it} (\log r) r^{it}) d\sigma(t). \quad (31)$$

Every polynomially weighted variation of σ is finite. The two logarithms cost $O(\log X) = X^{o(1)}$ on the terminal support.

Corollary 2.3 (Exact product-ready determinant layers). *Each coordinate in equations (18) and (19) is an $X^{o(1)}$ -projective superposition of layers of the form*

$$q_\omega(\gamma) \mathbf{1}_{j \equiv a \pmod{H_0}} \sum_{\substack{p, r, m \\ pr - mj = h_0}} a_\omega(p) b_\omega(r) c_\omega(m) \phi_\omega(j/J), \quad (32)$$

where the prime restriction is in a_ω , the factor $\mu(r)$ is in b_ω , and the literal source and mask coefficient is in c_ω . Equivalently, after opening $m = \ell d$, the layers have the form

$$\mathbf{1}_{j \equiv a \pmod{H_0}} \sum_{\substack{p, r, \ell, d \\ pr - d\ell = h_0}} a_\omega(p) b_\omega(r) c_\omega(\ell) f_\omega(d) \phi_\omega(j/J). \quad (33)$$

No arithmetic coefficient has been replaced by a smooth cutoff in either identity.

The residue selector is fixed because h_0 is fixed, but it is not a smooth function of j/J . It has the exact finite Fourier expansion

$$\mathbf{1}_{j \equiv a \pmod{H_0}} = \frac{1}{H_0} \sum_{\tau=0}^{H_0-1} e\left(\frac{\tau(j-a)}{H_0}\right). \quad (34)$$

Thus every product-ready layer is a finite, $O_{h_0}(1)$ -mass superposition of fixed modulations of a smooth orbit profile. It is *progression-smooth*, not literally an unmodulated smooth sequence. The progression form of the shift-spectral theorem is proved in [corollary 6.3](#): for a band-limited profile its exponent range is unchanged because H_0 is fixed. The selector alone does not make the actual profile band-limited, and it does not regularize the pulled-back dual field below.

This corollary is an interface theorem, not a scalar fixed-shift estimate. In particular, the coherent-complementary-factor theorem in TPC-36 leaves a character-level quantity of the form

$$\left| \sum_r w(r) \hat{R}_r(\chi) \right|^2 \quad (35)$$

and proves that Cauchy–Schwarz in the coherent variable can be sharp. Thus product-ready separation does not itself close even the scalar fixed- h_0 determinant sum.

2.3 Residual duality and the tensor coefficient

At the physical equality alias, the terminal support has $(d_\rho, n_{\rho,j}) = 1$. If $n_{\rho,j} = pr$ with $p > \sqrt{N_+}$, then d_ρ, p, r are pairwise coprime and the large-prime deletion leaves the residual label

$$s = d_\rho r. \quad (36)$$

After deleting zero atoms, every occurring r is squarefree; since $(d_\rho, r) = 1$ and d_ρ is squarefree, every occurring residual label $s = d_\rho r$ is squarefree. Let $\mathcal{H}_{\text{res}}$ be the residual Hilbert space with coordinates (i, s) , where $i \in \mathcal{I}_L \sqcup \mathcal{I}_R$ and s ranges over these supported squarefree labels. In the notation of TPC-45, define $x_p \in \mathcal{H}_{\text{res}}$ on the present large-prime sector by the same coordinate rule: its (i, s) -entry is the stripped atom with full label sp , and is zero when that atom is absent. This definition does not require the more restrictive very-high cutoff used for the singleton-incidence theorem there. Let

$$(Uy)_{i,s} = \mu(s)y_{i,s}. \quad (37)$$

Thus $|\mu(s)| = 1$ on the coordinate set, so U is unitary, and $\|\sum_p x_p\| = \|\sum_p Ux_p\|$.

Proposition 2.4 (Exact pulled-back dual tensor). *For $z = (z^L, z^R) \in \mathcal{H}_{\text{res}}$, the left part of the dual pairing is*

$$\left\langle \sum_{p \in \mathcal{P}_{R_0}^\#} Ux_p, z \right\rangle_L = - \sum_{\substack{\gamma, j, \rho = (\ell, d) \in \mathcal{A}_1, \\ p \in \mathcal{P}_{R_0}^\#, r \\ pr - dj\ell = h_0}} \gamma_\rho^{(1)} \mathfrak{A}_{\rho, \gamma}^{\text{act}}(j) \Lambda(pr) \mu(r) \overline{z_{\gamma, j, dr}^L}. \quad (38)$$

The right part is

$$\left\langle \sum_{p \in \mathcal{P}_{R_0}^\#} Ux_p, z \right\rangle_R = - \sum_{\substack{\alpha, j, \rho = (\ell, d) \in \mathcal{A}_2, \\ p \in \mathcal{P}_{R_0}^\#, r \\ pr - dj\ell = h_0}} \gamma_\rho^{(2)} \mathfrak{A}_{\alpha, \rho}^{\text{act}}(j) \Lambda(pr) \mu(r) \overline{z_{\alpha, j, dr}^R}. \quad (39)$$

Consequently, after the projective separation in [corollary 2.3](#), the new multiplier is precisely

$$q_\omega(\gamma)\phi_\omega(j/J)\mathbf{1}_{j\equiv a \pmod{H_0}}\overline{z_{\gamma,j,d(m)r}^L} \quad (40)$$

on the left, with the symmetric right-hand expression. It is a joint coefficient in the output row, j , the source divisor $d(m)$, and the cofactor r .

Proof. For a left atom with $n = pr$, TPC-45 gives the stripped coefficient

$$B_{\rho;\gamma,j}^L = -\gamma_\rho^{\circ,(1)}\mathfrak{A}_{\rho,\gamma}^{\text{act}}(j)\Lambda(pr). \quad (41)$$

The residual coordinate is $s = dr$. Multiplication by U inserts $\mu(s) = \mu(d)\mu(r)$. Combining $\mu(d)$ with $\gamma_\rho^{\circ,(1)}$ gives $\gamma_\rho^{(1)}$, and pairing with z gives (38). The right calculation is identical after exchanging the two row systems. Substitution of (26) yields (40). $\square$

The distinction from the scalar layer (32) is exact. An arbitrary unit vector z need not factor in (γ, j, dr) . Therefore (40) cannot be absorbed into the r -slot, the m -slot, or the smooth j -slot using the projective theorem of TPC-36. Expanding z into coordinate delta functions gives only a crude illustration of the possible cost. For one fixed opposite row there are at most

$$X^{o(1)}QJ = X^{1+o(1)} \quad (42)$$

active (j, s) -coordinates, so this expansion may cost $X^{1/2+o(1)}\|z\|_2$. Globally the number of atomic coordinates is at most

$$X^{o(1)}Q^2J = X^{667/400+o(1)}, \quad (43)$$

whose square-root scale is $X^{667/800+o(1)}$. These are crude upper bounds for one decomposition, not lower bounds or impossibility theorems. They show only that no $X^{o(1)}$ tensor lift follows from the inherited projective norm and $\|z\|_2 = 1$. The desired allowance is merely $K_B = X^{1/400+o(1)}$.

2.4 Three shift families

We finish by fixing terminology used later.

Definition 2.5 (Inherited completed-alias family). Let M_{al} be the triple-prime completion modulus of TPC-42 and put

$$c_k = h_0 + kM_{\text{al}}, \quad k \in \mathcal{I}_B. \quad (44)$$

The completed-alias column replaces $n_{\rho,j}$ by $n_{\rho,j,k} = m_\rho j + c_k$, with zero extension off the terminal support, while retaining the same row systems and the same coefficient $\gamma_\rho^{(s)}\mathfrak{A}^{\text{act}}(j)$ for every k .

The coefficient independence in this definition is an inherited fact, not an assumption [5, 6]. However, for $k \neq 0$ one may have

$$g_{\rho,j,k} = (d_\rho, n_{\rho,j,k}) = (d_\rho, c_k) > 1. \quad (45)$$

If $n_{\rho,j,k} = pr$ is squarefree and $p > d_\rho$, its fused Walsh label is

$$p \text{ sf}(d_\rho r), \quad \text{sf}(d_\rho r) = \frac{d_\rho r}{(d_\rho, r)^2}, \quad (46)$$

not necessarily $pd_\rho r$. Accordingly, the alias-averaged dual coefficient is

$$z_{k,\gamma,j,\text{sf}(dr)}^L \quad (47)$$

rather than the equality-alias coefficient $z_{\gamma,j,dr}^L$.

Definition 2.6 (Frozen completed family). For an auxiliary integer c , the frozen completed column keeps the physical h_0 row systems and source–mask coefficient field, but replaces the determinant equation by

$$pr - mj = c. \quad (48)$$

All terminal factors are evaluated at $n = pr = mj + c$ and are zero off support.

The inherited aliases $c = c_k$ are instances of this frozen family. For a general c , it is an algebraic coefficient model. It is not the TPC packet that would be obtained by restarting the construction with physical shift c .

Definition 2.7 (Actual local-shift family). For each nonzero shift ν , the actual local-shift packet is the packet reconstructed with physical shift ν . Its eligible rows, fixed residue factor $\Xi^{(\nu)}$, source residue data, terminal support, and joint multiplier are allowed to depend on ν .

TPC–36 proves $X^{o(1)}$ projective separation for each fixed ν , with a constant depending on ν . It does not prove a uniform bound over a growing local-shift window. Thus later spectral identities for the frozen family become statements about actual shifted TPC packets only after a separate uniform-provenance theorem. No such theorem is assumed in this paper.

3 A completed determinant model

The analytic statements are most transparent for a completed family in which the determinant value ranges over every integer. This section defines that family before any TPC specialization.

Let $\mathcal{P}, \mathcal{R}, \mathcal{M}, \mathcal{J} \subset \mathbb{N}$ be finite sets. The letter p is only an index in the abstract model; primality will be imposed in the TPC application. For coefficient sequences a, b, c, η , define

$$\mathcal{T}_h(a, b, c, \eta) := \sum_{\substack{p \in \mathcal{P}, r \in \mathcal{R}, m \in \mathcal{M}, j \in \mathcal{J} \\ pr - mj = h}} a_p b_r c_m \eta_j, \quad h \in \mathbb{Z}. \quad (49)$$

The corresponding Cartesian tensor energy and fixed-shift atomic diagonal are

$$\mathcal{D}_\otimes := \left(\sum_{p,r} |a_p b_r|^2 \right) \left(\sum_{m,j} |c_m \eta_j|^2 \right), \quad (50)$$

$$\Delta_h := \sum_{pr - mj = h} |a_p b_r c_m \eta_j|^2. \quad (51)$$

Every Cartesian tuple has one and only one determinant value, and hence

$$\boxed{\sum_{h \in \mathbb{Z}} \Delta_h = \mathcal{D}_\otimes.} \quad (52)$$

This elementary identity is the normalization divide on which several later conclusions turn.

For Hilbert-valued spectral statements it is useful to allow an arbitrary coefficient on the three arithmetic variables. Let $\mathcal{H}$ be a complex Hilbert space, let $z_{p,r,m} \in \mathbb{C}$, and let $v_{p,r,m} \in \mathcal{H}$. For a profile w on $\mathbb{R}$, put

$$F_h := \sum_{p,r,m} \sum_{j \in \mathbb{Z}} z_{p,r,m} w(j/J) v_{p,r,m} \mathbf{1}_{pr - mj = h}. \quad (53)$$

The arithmetic supports in p, r, m remain finite, while the Schwartz or compactly supported profile makes the j -sum absolutely convergent. The essential hypothesis in (53) is that the

Hilbert vector is independent of j . An arbitrary vector $v_{p,r,m,j}$ can encode a different orthogonal output at every orbit time and need not have any orbit spectral gap.

When a multiplicative residual label is retained, fix a map

$$d : \mathcal{M} \longrightarrow \mathbb{N} \quad (54)$$

and define

$$F(h, s) := \sum_{\substack{pr-mj=h \\ d(m)r=s}} a_p b_r c_m \eta_j. \quad (55)$$

At the physical TPC equality alias, $d(m)$ is the source divisor in the unique row $m = \ell d(m)$, and coprimality gives the residual label $s = d(m)r$. For nonzero completed aliases, the squarefree kernel may replace this literal product; that distinction is treated in [section 9](#).

Finally define the product-fiber multiplicities

$$\rho_{\mathcal{P}, \mathcal{R}} := \max_n \# \{ (p, r) \in \mathcal{P} \times \mathcal{R} : pr = n \}, \quad (56)$$

$$\rho_{\mathcal{M}, \mathcal{J}} := \max_n \# \{ (m, j) \in \mathcal{M} \times \mathcal{J} : mj = n \}. \quad (57)$$

For polynomially bounded integer supports,

$$\rho_{\mathcal{M}, \mathcal{J}} \leq \max_{n \leq X^{O(1)}} \tau(n) = X^{o(1)}. \quad (58)$$

At the optimized TPC endpoint the four principal lengths are

$$PR = X^{1+o(1)}, \quad M = X^{267/400+o(1)}, \quad J = X^{133/400+o(1)}, \quad MJ = X^{1+o(1)}. \quad (59)$$

The notation M in (59) is the source-variable length. The triple-prime alias modulus will always be written M_{al} to prevent a collision of notation.

4 Fixed-shift closure at tensor normalization

The third arithmetic coefficient is not, by itself, an obstruction to a fixed-shift estimate. The issue is the norm in which that estimate is obtained.

Define multiplicative synthesis sequences

$$A(n) := \sum_{pr=n} a_p b_r, \quad C(n) := \sum_{mj=n} c_m \eta_j. \quad (60)$$

Then

$$\mathcal{T}_h(a, b, c, \eta) = \sum_n A(n) C(n-h). \quad (61)$$

Theorem 4.1 (Scalar tensor closure). *For every integer h ,*

$$\boxed{|\mathcal{T}_h(a, b, c, \eta)|^2 \leq \rho_{\mathcal{P}, \mathcal{R}} \rho_{\mathcal{M}, \mathcal{J}} \mathcal{D}_{\otimes}}. \quad (62)$$

Proof. Cauchy–Schwarz in (61) gives $|\mathcal{T}_h|^2 \leq \|A\|_2^2 \|C\|_2^2$. On each product fiber, Cauchy–Schwarz gives

$$\begin{aligned} \|A\|_2^2 &\leq \rho_{\mathcal{P}, \mathcal{R}} \sum_{p,r} |a_p b_r|^2, \\ \|C\|_2^2 &\leq \rho_{\mathcal{M}, \mathcal{J}} \sum_{m,j} |c_m \eta_j|^2. \end{aligned}$$

Multiplication proves the claim. □

The same proof is valid with one synthesized sequence Hilbert-valued.

Corollary 4.2 (One-vector-slot closure). *Let $x_{p,r} \in \mathbb{C}$ and $y_{m,j} \in \mathcal{H}$, and put*

$$G_h := \sum_{pr-mj=h} x_{p,r} y_{m,j}.$$

Then

$$\|G_h\|_{\mathcal{H}}^2 \leq \rho_{\mathcal{P},\mathcal{R}} \rho_{\mathcal{M},\mathcal{J}} \left(\sum_{p,r} |x_{p,r}|^2 \right) \left(\sum_{m,j} \|y_{m,j}\|_{\mathcal{H}}^2 \right). \quad (63)$$

Proof. Use Hilbert-space Cauchy–Schwarz after product synthesis; the fiberwise inequality $\|\sum_{\nu=1}^k y_{\nu}\|^2 \leq k \sum_{\nu} \|y_{\nu}\|^2$ replaces its scalar counterpart. $\square$

In the global large-prime range of TPC–45, product synthesis on the left is actually injective.

Lemma 4.3 (Large-prime product injectivity). *Suppose every $p \in \mathcal{P}$ is prime, $p > \sqrt{N_+}$, and every occurring product satisfies $pr \leq N_+$. Then*

$$\rho_{\mathcal{P},\mathcal{R}} = 1. \quad (64)$$

Proof. If $pr = p'r' \leq N_+$ with distinct primes $p, p' > \sqrt{N_+}$, then pp' divides the common product, while $pp' > N_+$, a contradiction. $\square$

Combining [theorem 4.1](#) and [lemma 4.3](#) with (58) gives

$$|\mathcal{T}_h|^2 \leq X^{o(1)} \mathcal{D}_{\otimes} \quad (65)$$

uniformly in h . This estimate retains an arbitrary prime coefficient, an arbitrary cofactor coefficient, and an arbitrary source coefficient. It uses neither determinant-shift averaging nor a Kloosterman-fraction estimate.

The conclusion is nevertheless not the TPC–45 Bessel target. By (52), the right side of (65) is the total atomic energy over *all* determinant shifts. TPC–45 requires the atomic diagonal at the one physical shift and inside the actual output geometry. The size and sharpness of that mismatch are quantified in [section 8](#).

5 Fourier–Bohr scalarization of the residual label

The multiplicative label $s = d(m)r$ can be retained without paying a dimension factor. This shows that the bare residual product is not the source of the tensor-normalization loss.

Let $\mathcal{Q}$ be a finite prime set containing all prime factors of every r and $d(m)$ in the packet, and put

$$\mathbb{T}^{\mathcal{Q}} := \prod_{q \in \mathcal{Q}} \{z_q \in \mathbb{C} : |z_q| = 1\} \quad (66)$$

with normalized Haar measure dz . For $z = (z_q)_{q \in \mathcal{Q}}$, define the completely multiplicative character

$$\chi_z(n) := \prod_{q \in \mathcal{Q}} z_q^{v_q(n)}. \quad (67)$$

Orthogonality gives

$$\int_{\mathbb{T}^{\mathcal{Q}}} \chi_z(s) \overline{\chi_z(t)} dz = \mathbf{1}_{s=t} \quad (68)$$

for the supported integers s, t .

Theorem 5.1 (Residual-label fixed-shift closure). *For the residual packet in (55),*

$$\boxed{\sum_s |F(h, s)|^2 \leq \rho_{\mathcal{P}, \mathcal{R}} \rho_{\mathcal{M}, \mathcal{J}} \mathcal{D}_{\otimes}.} \quad (69)$$

Proof. The Bohr transform of $F(h, \cdot)$ is

$$\begin{aligned} \tilde{F}_h(z) &:= \sum_s F(h, s) \chi_z(s) \\ &= \sum_{pr-mj=h} a_p b_r c_m \eta_j \chi_z(r) \chi_z(d(m)). \end{aligned} \quad (70)$$

For each fixed z , this is the scalar determinant form with unit-modulus twists absorbed into b_r and c_m . Theorem 4.1 therefore bounds $|\tilde{F}_h(z)|^2$ by the right side of (69), uniformly in z . Parseval in (68) and integration over z finish the proof. $\square$

There is also a two-spectrum identity which records every remaining collision exactly. Put

$$A_z(\alpha) := \sum_{p,r} a_p b_r \chi_z(r) e(-pr\alpha), \quad (71)$$

$$C_z(\alpha) := \sum_{m,j} c_m \eta_j \chi_z(d(m)) e(mj\alpha). \quad (72)$$

Theorem 5.2 (Additive–multiplicative Parseval normal form). *One has the exact identity*

$$\boxed{\sum_{h,s} |F(h, s)|^2 = \int_{\mathbb{T}} \int_{\mathbb{T}^{\mathcal{Q}}} |A_z(\alpha)|^2 |C_z(\alpha)|^2 dz d\alpha.} \quad (73)$$

Proof. The joint Fourier–Bohr transform of $F(h, s)$ equals

$$\begin{aligned} \sum_{h,s} F(h, s) \chi_z(s) e(-h\alpha) &= \sum_{p,r,m,j} a_p b_r c_m \eta_j \chi_z(r) \chi_z(d(m)) e(-(pr - mj)\alpha) \\ &= A_z(\alpha) C_z(\alpha). \end{aligned}$$

Parseval on $\mathbb{T} \times \mathbb{T}^{\mathcal{Q}}$ gives (73). $\square$

Expansion of (73) shows that its off-diagonal terms are exactly the pairs satisfying both

$$pr - mj = p'r' - m'j', \quad d(m)r = d(m')r'. \quad (74)$$

Thus the remaining Gram problem is a mixed additive–multiplicative energy, not an unspecified appeal to Möbius randomness.

For squarefree-kernel labels, the same argument uses the finite Walsh group $\{\pm 1\}^{\mathcal{Q}}$ and prime valuations modulo two. At the physical equality alias, TPC coprimality makes the simpler product version above literal.

6 The shift spectrum of a smooth determinant packet

We study the completed Hilbert packet (53). This is a Fourier model for the equation $pr - mj = h$; it is not yet an assertion that the actual TPC packet has been constructed uniformly for every h . Let $J \geq 2$ and suppose that its source support satisfies

$$0 < M_- \leq m \leq M_+,$$

The coefficients and vectors have finite arithmetic support and $w \in \mathcal{S}(\mathbb{R})$. The vectors in (53) may depend jointly and arbitrarily on (p, r, m) , but they do not depend on j . In particular, the residual-label choice

$$v_{p,r,m} = e_{d(m)r} \quad \text{in } \ell^2(\mathbb{N}) \quad (75)$$

is permitted. The sequence $(F_h)_h$ belongs to both $\ell^1(\mathbb{Z}; \mathcal{H})$ and $\ell^2(\mathbb{Z}; \mathcal{H})$. We use the shift Fourier transform

$$\widehat{F}(\alpha) := \sum_{h \in \mathbb{Z}} F_h e(-h\alpha), \quad \alpha \in \mathbb{T}. \quad (76)$$

Theorem 6.1 (Hilbert-valued orbit kernel). *For every $\alpha \in \mathbb{T}$,*

$$\widehat{F}(\alpha) = \sum_{p,r,m} z_{p,r,m} v_{p,r,m} e(-pr\alpha) K_m(\alpha), \quad (77)$$

$$K_m(\alpha) := \sum_{j \in \mathbb{Z}} w(j/J) e(mj\alpha) = J \sum_{k \in \mathbb{Z}} \widehat{w}(J(k - m\alpha)). \quad (78)$$

Moreover,

$$\sum_{h \in \mathbb{Z}} \|F_h\|_{\mathcal{H}}^2 = \int_{\mathbb{T}} \|\widehat{F}(\alpha)\|_{\mathcal{H}}^2 d\alpha. \quad (79)$$

Proof. Absolute convergence allows the order of summation to be changed. Substituting $h = pr - mj$ into (76) gives the first identity. Poisson summation applied to $x \mapsto w(x/J)$ gives

$$\sum_{j \in \mathbb{Z}} w(j/J) e(mj\alpha) = \sum_{k \in \mathbb{Z}} J \widehat{w}(J(k - m\alpha)),$$

which proves the second. The last assertion is the standard Hilbert-valued Parseval identity; it also follows by applying scalar Parseval after pairing with an orthonormal basis of the finite dimensional span of the vectors $v_{p,r,m}$. $\square$

The factor K_m contains all shift frequencies contributed by the smooth orbit variable. Neither the primality of p nor any arithmetic cancellation in the three coefficient slots is used in the following theorem.

Theorem 6.2 (Exact mesoscopic shift-spectral gap). *Assume that, for some $\sigma > 0$,*

$$\text{supp } \widehat{w} \subset [-\sigma, \sigma] \quad \text{and} \quad J > \sigma. \quad (80)$$

Then

$$\widehat{F}(\alpha) = 0 \quad (81)$$

whenever

$$\boxed{\frac{\sigma}{M_- J} < \|\alpha\|_{\mathbb{T}} < \frac{1 - \sigma/J}{M_+}}. \quad (82)$$

The assertion is vacuous if the displayed interval is empty.

Proof. Formula (78) and (80) show that

$$K_m(\alpha) \neq 0 \implies \|m\alpha\|_{\mathbb{T}} \leq \frac{\sigma}{J}. \quad (83)$$

Put $t = \|\alpha\|_{\mathbb{T}}$ and choose a representative with $|\alpha| = t$. Under (82), every $m \in [M_-, M_+]$ satisfies

$$\frac{\sigma}{J} < mt < 1 - \frac{\sigma}{J}.$$

Thus the distance of $m\alpha$ from both 0 and the nearest nonzero integer is greater than σ/J . This contradicts (83), so every $K_m(\alpha)$ in (77) vanishes. $\square$

Corollary 6.3 (Fixed-progression shift gap). *Let $H_0 \in \mathbb{N}$, let a be a residue class modulo H_0 , and replace (53) by*

$$F_h^{a, H_0} := \sum_{p, r, m} \sum_{\substack{j \in \mathbb{Z} \\ j \equiv a \pmod{H_0}}} z_{p, r, m} w(j/J) v_{p, r, m} \mathbf{1}_{pr - mj = h}. \quad (84)$$

Assume $\text{supp } \hat{w} \subset [-\sigma, \sigma]$ and $J > H_0 \sigma$. Then

$$\hat{F}^{a, H_0}(\alpha) = 0 \quad (85)$$

whenever

$$\boxed{\frac{\sigma}{M_- J} < \|\alpha\|_{\mathbb{T}} < \frac{H_0^{-1} - \sigma/J}{M_+}}. \quad (86)$$

For fixed H_0 , the endpoint exponent range remains $X^{-1+o(1)} < \|\alpha\|_{\mathbb{T}} < X^{-267/400+o(1)}$.

Proof. The progression-restricted orbit kernel is

$$\begin{aligned} K_{m, a}^{(H_0)}(\alpha) &:= \sum_{\substack{j \in \mathbb{Z} \\ j \equiv a \pmod{H_0}}} w(j/J) e(mj\alpha) \\ &= \frac{J}{H_0} \sum_{k \in \mathbb{Z}} e(ak/H_0) \hat{w}\left(\frac{J}{H_0}(k - mH_0\alpha)\right). \end{aligned} \quad (87)$$

Indeed, write $j = a + H_0 t$ and apply Poisson summation in t . Thus a nonzero kernel requires

$$\|mH_0\alpha\|_{\mathbb{T}} \leq \frac{H_0\sigma}{J}.$$

If $t = \|\alpha\|_{\mathbb{T}}$ lies in (86), then every supported m satisfies

$$\frac{H_0\sigma}{J} < mH_0t < 1 - \frac{H_0\sigma}{J}.$$

The same support argument as in [theorem 6.2](#) makes every $K_{m, a}^{(H_0)}(\alpha)$ vanish. The endpoint statement follows because H_0 is fixed. $\square$

Corollary 6.4 (The endpoint spectral no-man's-land). *Suppose*

$$M_- = X^{267/400+o(1)}, \quad M_+ = X^{267/400+o(1)}, \quad J = X^{133/400+o(1)}. \quad (88)$$

For a fixed band limit σ , the exact gap in [theorem 6.2](#) has the exponent form

$$X^{-1+o(1)} < \|\alpha\|_{\mathbb{T}} < X^{-267/400+o(1)}. \quad (89)$$

Thus the zero-frequency orbit envelope, whose width is of order $(MJ)^{-1}$, is separated from the first lattice aliases, whose distance from zero is of order M^{-1} .

Strict Fourier compactness is convenient for an exact statement, but is not needed for a power-robust gap.

Theorem 6.5 (Rapid decay for a smooth compact orbit profile). *Let $w \in C_c^\infty(\mathbb{R})$, and suppose that for fixed $\mu, \nu > 0$ with $\mu + \nu = 1$,*

$$M_- \asymp M_+ \asymp X^\mu, \quad J \asymp X^\nu. \quad (90)$$

Fix $0 < \epsilon < \nu$. For every $A > 0$, uniformly on

$$X^{-1+\epsilon} \leq \|\alpha\|_{\mathbb{T}} \leq X^{-\mu-\epsilon}, \quad (91)$$

one has

$$|K_m(\alpha)| \ll_{A,w,\epsilon} JX^{-A\epsilon} \quad (M_- \leq m \leq M_+), \quad (92)$$

$$\|\widehat{F}(\alpha)\|_{\mathcal{H}} \ll_{A,w,\epsilon} JX^{-A\epsilon} \sum_{p,r,m} |z_{p,r,m}| \|v_{p,r,m}\|_{\mathcal{H}}. \quad (93)$$

Replacing A by a larger constant absorbs any fixed polynomial size of the coefficient support.

Proof. Let $t = \|\alpha\|_{\mathbb{T}}$. Uniformly in the range (91),

$$Jmt \gg X^\epsilon, \quad mt \ll X^{-\epsilon}.$$

For all sufficiently large X , the nearest integer to $m\alpha$ is therefore zero. Since $\widehat{w}$ is Schwartz,

$$J|\widehat{w}(Jm\alpha)| \ll_{A,w} J(Jmt)^{-A} \ll_{A,w,\epsilon} JX^{-A\epsilon}.$$

For $k \neq 0$, one has $|k - m\alpha| \geq |k|/2$. Given the requested output exponent A , choose a Schwartz decay order B with

$$B \geq A, \quad B\nu \geq A\epsilon.$$

Then

$$J \sum_{k \neq 0} |\widehat{w}(J(k - m\alpha))| \ll_{B,w} J^{1-B} \ll_{A,w,\epsilon} JX^{-A\epsilon}.$$

The zero mode is bounded with the same order B , and hence also by the right-hand side of (92). The triangle inequality in (77) proves (93). $\square$

Remark 6.6 (Why the vectors must be orbit independent). If $v_{p,r,m}$ in (53) is replaced by an arbitrary $v_{p,r,m,j}$, then $K_m(\alpha)$ is replaced by the vector-valued discrete transform

$$\sum_j w(j/J) v_{p,r,m,j} e(mj\alpha),$$

which may have full shift spectrum. The preceding theorems continue to hold under a separately proved uniform band-limit or rapid Fourier decay for these vector-valued profiles; they do not hold for arbitrary orbit-output coupling. In particular, no smoothness of an actual TPC projective mask is being assumed implicitly here.

Remark 6.7 (Amplitude gap versus energy). The vanishing in (81) concerns the Fourier transform of the Hilbert-valued amplitude F_h . It gives an orthogonal low-frequency/high-alias decomposition of the total Parseval energy (79). It does not give the same spectral gap for the scalar sequence $h \mapsto \|F_h\|_{\mathcal{H}}^2$: multiplication in h convolves shift frequencies, and differences of low and alias frequencies may occupy the intermediate band. Consequently, the amplitude gap alone is not an atomic-normalized Bessel estimate at a prescribed shift.

7 Poisson formulas for coarse shift averages

We now test the completed packet (53) by a window of physical shift length $H > 0$. For $h_0 \in \mathbb{R}$ and $W \in \mathcal{S}(\mathbb{R})$, put

$$\mathcal{A}_{H,h_0}(W; F) := \frac{1}{H} \sum_{h \in \mathbb{Z}} W\left(\frac{h - h_0}{H}\right) F_h. \quad (94)$$

All identities below are Bochner identities in $\mathcal{H}$. Their proofs are coefficient blind; the vectors $v_{p,r,m}$ remain arbitrary in (p, r, m) , but, as in section 6, they must be independent of j .

7.1 An exact band-pass null interval

Theorem 7.1 (Exact band-pass vanishing). *Assume*

$$\text{supp } \widehat{w} \subset [-\sigma, \sigma], \quad J > \sigma, \quad (95)$$

and let $\psi \in \mathcal{S}(\mathbb{R})$ satisfy

$$\text{supp } \widehat{\psi} \subset \{\xi \in \mathbb{R} : a \leq |\xi| \leq b\} \quad (0 < a \leq b). \quad (96)$$

If

$$\boxed{\frac{bM_+}{1 - \sigma/J} < H < \frac{aM_-J}{\sigma}}, \quad (97)$$

then, for every h_0 ,

$$\boxed{\mathcal{A}_{H,h_0}(\psi; F) = 0}. \quad (98)$$

Proof. Insert (53) into (94). For a fixed triple (p, r, m) , the remaining scalar sum is

$$\frac{1}{H} \sum_{j \in \mathbb{Z}} f_{p,r,m}(j), \quad f_{p,r,m}(x) := w(x/J) \psi\left(\frac{pr - h_0 - mx}{H}\right). \quad (99)$$

The Fourier support of $x \mapsto w(x/J)$ is contained in $[-\sigma/J, \sigma/J]$. The Fourier support of the second factor in (99) is contained in

$$[-bm/H, -am/H] \cup [am/H, bm/H].$$

Consequently, every point of $\text{supp } \widehat{f}_{p,r,m}$ has absolute value between

$$\frac{am}{H} - \frac{\sigma}{J} \quad \text{and} \quad \frac{bm}{H} + \frac{\sigma}{J}.$$

The upper inequality in (97) makes the lower endpoint positive for all $m \geq M_-$, while the lower inequality makes the upper endpoint smaller than one for all $m \leq M_+$. Hence the Fourier support contains no integer. Poisson summation gives $\sum_j f_{p,r,m}(j) = \sum_k \widehat{f}_{p,r,m}(k) = 0$. Summing over the finite arithmetic support proves the theorem. $\square$

At the endpoint (88), fixed band limits give a nonempty physical window range

$$X^{267/400+o(1)} < H < X^{1-o(1)}. \quad (100)$$

This is a statement about signed band-pass averages of the completed amplitude. It is not a positive average of its squared norm.

7.2 The exact low-pass main term

Theorem 7.2 (Exact low-pass Poisson formula). *Assume (95), and let $W \in \mathcal{S}(\mathbb{R})$ satisfy*

$$\text{supp } \widehat{W} \subset [-b, b]. \quad (101)$$

If

$$\frac{bM_+}{H} + \frac{\sigma}{J} < 1, \quad (102)$$

then

$$\mathcal{A}_{H,h_0}(W; F) = \sum_{p,r,m} \frac{z_{p,r,m}}{m} v_{p,r,m} \int_{\mathbb{R}} W(u) w\left(\frac{pr - h_0 - Hu}{mJ}\right) du. \quad (103)$$

Proof. For fixed (p, r, m) , set

$$f(x) := w(x/J)W\left(\frac{pr - h_0 - mx}{H}\right).$$

The Fourier support of f is contained in

$$\left[-\frac{\sigma}{J} - \frac{bm}{H}, \frac{\sigma}{J} + \frac{bm}{H}\right],$$

which lies strictly inside $(-1, 1)$ by (102). Poisson summation thus has only its zero mode:

$$\sum_{j \in \mathbb{Z}} f(j) = \hat{f}(0) = \int_{\mathbb{R}} f(x) dx.$$

With $u = (pr - h_0 - mx)/H$,

$$\frac{1}{H} \int_{\mathbb{R}} f(x) dx = \frac{1}{m} \int_{\mathbb{R}} W(u)w\left(\frac{pr - h_0 - Hu}{mJ}\right) du.$$

Multiplying by $z_{p,r,m}v_{p,r,m}$ and summing proves (103). $\square$

For later comparison, define the frozen-envelope main term

$$\mathcal{M}_{H,h_0}(W; F) := \left(\int_{\mathbb{R}} W(u) du \right) \sum_{p,r,m} \frac{z_{p,r,m}}{m} v_{p,r,m} w\left(\frac{pr - h_0}{mJ}\right). \quad (104)$$

Corollary 7.3 (Variation of the exact low-pass main). *Under the hypotheses of theorem 7.2,*

$$\|\mathcal{A}_{H,h_0}(W; F) - \mathcal{M}_{H,h_0}(W; F)\|_{\mathcal{H}} \ll_{W,w} \frac{H}{M_- J} \sum_{p,r,m} \frac{|z_{p,r,m}|}{m} \|v_{p,r,m}\|_{\mathcal{H}}. \quad (105)$$

Proof. The mean-value theorem gives, uniformly in $m \geq M_-$,

$$\left| w\left(\frac{pr - h_0 - Hu}{mJ}\right) - w\left(\frac{pr - h_0}{mJ}\right) \right| \leq \|w'\|_{\infty} \frac{H|u|}{M_- J}.$$

Apply this in (103), integrate $|uW(u)|$, and use the triangle inequality in $\mathcal{H}$. $\square$

The same main term survives for ordinary compactly supported smooth profiles, with a rapidly decreasing sampling error.

Theorem 7.4 (Smooth Poisson approximation). *Let $W, w \in C_c^{\infty}(\mathbb{R})$. Suppose*

$$H \geq 2M_+, \quad H \leq C_0 M_- J \quad (106)$$

for a fixed C_0 . For every $A > 0$,

$$\begin{aligned} \mathcal{A}_{H,h_0}(W; F) &= \sum_{p,r,m} \frac{z_{p,r,m}}{m} v_{p,r,m} \\ &\quad \times \int_{\mathbb{R}} W(u)w\left(\frac{pr - h_0 - Hu}{mJ}\right) du + \mathcal{E}_{H,h_0}, \end{aligned} \quad (107)$$

$$\begin{aligned} \|\mathcal{E}_{H,h_0}\|_{\mathcal{H}} &\ll_{A,W,w,C_0} \left(\frac{M_+}{H}\right)^A \\ &\quad \times \sum_{p,r,m} \frac{|z_{p,r,m}|}{m} \|v_{p,r,m}\|_{\mathcal{H}}. \end{aligned} \quad (108)$$

Consequently,

$$\begin{aligned} \|\mathcal{A}_{H,h_0}(W;F) - \mathcal{M}_{H,h_0}(W;F)\|_{\mathcal{H}} &\ll \left[\left(\frac{M_+}{H} \right)^A + \frac{H}{M_-J} \right] \\ &\times \sum_{p,r,m} \frac{|z_{p,r,m}|}{m} \|v_{p,r,m}\|_{\mathcal{H}}, \end{aligned} \quad (109)$$

where the implicit constant has the same fixed dependencies.

Proof. For one triple put $B = pr - h_0$ and

$$f(x) = w(x/J)W((B - mx)/H).$$

Poisson summation writes $\sum_j f(j) = \sum_k \widehat{f}(k)$. After the substitution $u = (B - mx)/H$,

$$\widehat{f}(k) = \frac{H}{m} e(-kB/m) \int_{\mathbb{R}} W(u) w\left(\frac{B - Hu}{mJ}\right) e(kHu/m) du. \quad (110)$$

The $k = 0$ term gives the integral in (107). Set $\delta = H/m$. The range (106) gives $\delta \geq 2$ and $H/(mJ) \leq C_0$. Repeated integration by parts in (110) therefore yields, for $k \neq 0$,

$$|\widehat{f}(k)| \ll_{A,W,w,C_0} \delta(|k|\delta)^{-A-1}.$$

After summing $k \neq 0$ and dividing by $H = m\delta$, the error for this triple is

$$\ll_{A,W,w,C_0} \frac{1}{m} \left(\frac{m}{H} \right)^A \leq \frac{1}{m} \left(\frac{M_+}{H} \right)^A.$$

The Hilbert-space triangle inequality proves (108). The mean-value argument from [corollary 7.3](#) then gives (109). $\square$

At the TPC scales (88), taking $H = X^{267/400+\epsilon}$ with $0 < \epsilon < 133/400$ makes the two errors in (109)

$$X^{-A\epsilon+o(1)} \quad \text{and} \quad X^{-133/400+\epsilon+o(1)}, \quad (111)$$

respectively. This strong approximation concerns a coarse completed shift average, not the value at a prescribed physical shift.

7.3 The averaged atomic diagonal

The diagonal associated with (53) is

$$\Delta_h^F := \sum_{p,r,m} |z_{p,r,m}|^2 \|v_{p,r,m}\|_{\mathcal{H}}^2 \sum_{j \in \mathbb{Z}} |w(j/J)|^2 \mathbf{1}_{\{h=pr-mj\}}. \quad (112)$$

It contains exactly the equal-atom terms in $\|F_h\|_{\mathcal{H}}^2$, not the cross terms between distinct determinant atoms.

Theorem 7.5 (Exact averaged atomic diagonal). *Assume (95) and (101). If*

$$\frac{bM_+}{H} + \frac{2\sigma}{J} < 1, \quad (113)$$

then

$$\begin{aligned} \frac{1}{H} \sum_{h \in \mathbb{Z}} W\left(\frac{h - h_0}{H}\right) \Delta_h^F &= \sum_{p,r,m} \frac{|z_{p,r,m}|^2}{m} \|v_{p,r,m}\|_{\mathcal{H}}^2 \\ &\times \int_{\mathbb{R}} W(u) \left| w\left(\frac{pr - h_0 - Hu}{mJ}\right) \right|^2 du. \end{aligned} \quad (114)$$

If $W, w \in C_c^\infty(\mathbb{R})$ instead, under (106) one has, for every $A > 0$,

$$\begin{aligned} & \frac{1}{H} \sum_h W\left(\frac{h-h_0}{H}\right) \Delta_h^F \\ &= \left(\int_{\mathbb{R}} W\right) \sum_{p,r,m} \frac{|z_{p,r,m}|^2}{m} \|v_{p,r,m}\|_{\mathcal{H}}^2 \left|w\left(\frac{pr-h_0}{mJ}\right)\right|^2 + \mathcal{E}_\Delta, \end{aligned} \quad (115)$$

$$|\mathcal{E}_\Delta| \ll_{A,W,w,C_0} \left[\left(\frac{M_+}{H}\right)^A + \frac{H}{M_-J} \right] \sum_{p,r,m} \frac{|z_{p,r,m}|^2}{m} \|v_{p,r,m}\|_{\mathcal{H}}^2. \quad (116)$$

Proof. Insert (112) and eliminate h . The resulting j -sum is the one in theorem 7.2 with w replaced by $|w|^2$. Since

$$\text{supp } \widehat{|w|^2} \subset [-2\sigma, 2\sigma],$$

condition (113) leaves only the zero Poisson mode and proves (114). For smooth compactly supported profiles, repeat the proof of theorem 7.4 with $|w|^2$ in place of w , and then use the mean-value theorem. This gives (115) and (116). $\square$

Remark 7.6 (What the diagonal formula does not estimate). In general,

$$\|F_h\|_{\mathcal{H}}^2 = \Delta_h^F + \sum_{\substack{e \neq e' \\ h(e)=h(e')}} \langle b_e, b_{e'} \rangle_{\mathcal{H}},$$

where e, e' denote determinant atoms. Theorems 7.2 and 7.5 compute the averaged amplitude and its atomic diagonal, respectively; neither deletes the equal-determinant cross terms in the averaged squared norm. Moreover, the signed band-pass identity (98) cannot be squared and reinterpreted as a positive energy estimate. These distinctions are essential when comparing the completed shift model with the atomic-normalized TPC Gram/Bessel gate.

8 Sharp atomic-normalization boundaries

This section shows why the valid tensor estimates in sections 4 and 5 cannot be substituted for the physical Bessel estimate.

8.1 Averaging all shifts can retain a polynomial loss

Take nonnegative unit coefficients on dyadic boxes and let

$$N_h := \#\{(p, r, m, j) : pr - mj = h\}. \quad (117)$$

Then $\mathcal{T}_h = N_h = \Delta_h$. Suppose all differences occupy an interval of $O(X)$ integers. If

$$K_1 := \#(\mathcal{P} \times \mathcal{R}), \quad K_2 := \#(\mathcal{M} \times \mathcal{J}), \quad (118)$$

then $\sum_h N_h = K_1 K_2$, and Cauchy-Schwarz gives

$$\sum_h |\mathcal{T}_h|^2 = \sum_h N_h^2 \gg \frac{(K_1 K_2)^2}{X}. \quad (119)$$

Since $\sum_h \Delta_h = K_1 K_2$,

$$\frac{\sum_h |\mathcal{T}_h|^2}{\sum_h \Delta_h} \gg \frac{K_1 K_2}{X}. \quad (120)$$

At the endpoint, a prime p -box and three integer boxes have

$$K_1 = X^{1+o(1)}/\log X, \quad K_2 = X^{1+o(1)}, \quad (121)$$

so the ratio in (120) is

$$\gg X^{1-o(1)}. \quad (122)$$

Partitioning the difference range into intervals of any prescribed length $1 \leq H \ll X$ shows that at least one nonempty H -window has ratio at least the global ratio. Therefore no theorem for all coefficient boxes and all shift windows can replace tensor energy by the averaged atomic diagonal at subpolynomial cost.

8.2 A prescribed-shift obstruction for arbitrary slots

The following construction keeps the prime condition but is not asserted to satisfy the actual TPC source factorization or Möbius weights.

Proposition 8.1 (Fixed-shift coherent box). *Fix $h \neq 0$. Let $p \asymp P$ range over primes, choose $j_0 \in \mathbb{N}$ with $j_0 \asymp J$ and $2J < P$, and let $R \geq J$ be an integer. Work at a scale for which $pR > |h|$ for every selected prime, and choose the boxes so that $j_0 < \min p$ over those primes. For every such p , choose the unique*

$$r_p \in [R, R + j_0) \quad \text{with} \quad pr_p \equiv h \pmod{j_0}, \quad (123)$$

and put

$$m_p := \frac{pr_p - h}{j_0}. \quad (124)$$

Let a, b, c be the indicators of the resulting p, r_p, m_p sets, and let $0 \leq \eta_j \leq 1$ with $\eta_{j_0} = 1$. Then

$$\frac{|\mathcal{T}_h|^2}{\Delta_h} \geq \#\{p \asymp P : p \text{ prime}\} = P^{1-o(1)}. \quad (125)$$

Proof. Since $p > j_0$, the prime p is invertible modulo j_0 , so (123) has exactly one representative in the declared half-open interval. Every selected prime supplies the solution $pr_p - m_p j_0 = h$. The scale hypothesis gives $m_p > 0$. All terms are nonnegative. Since every coefficient has modulus at most one, $\Delta_h \leq \mathcal{T}_h$, while $\mathcal{T}_h$ is at least the number of selected primes. Hence $|\mathcal{T}_h|^2/\Delta_h \geq \mathcal{T}_h$, proving the claim. $\square$

At the TPC lengths $P = M = X^{267/400+o(1)}$ and $R = J = X^{133/400+o(1)}$, this abstract obstruction has exponent $267/400$, whereas the endpoint budget is $1/400$. The gap is

$$\frac{267}{400} - \frac{1}{400} = \frac{133}{200}. \quad (126)$$

The construction rules out an arbitrary-three-slot black box at atomic normalization. It does not say that the actual TPC mask realizes the coherent coefficients.

8.3 Weighted shift averaging and exact grouping norms

Let E be a finite atom set with maps $h : E \rightarrow \mathbb{Z}$ and $s : E \rightarrow S$. For nonnegative weights ω_h , define

$$\|Gx\|_\omega^2 := \sum_{h_0, s_0} \omega_{h_0} \left| \sum_{\substack{e: h(e)=h_0 \\ s(e)=s_0}} x_e \right|^2, \quad \|x\|_{\omega, E}^2 := \sum_e \omega_{h(e)} |x_e|^2. \quad (127)$$

Theorem 8.2 (Averaging-invariant grouping norm). *After discarding zero-weight fibers,*

$$\sup_{x \neq 0} \frac{\|Gx\|_\omega^2}{\|x\|_{\omega,E}^2} = \max_{\substack{h_0, s_0 \\ \omega_{h_0} > 0}} \#\{e : h(e) = h_0, s(e) = s_0\}. \quad (128)$$

Proof. Fiberwise Cauchy–Schwarz gives the upper bound. On a largest positive-weight fiber, take x constant and nonzero and set it to zero elsewhere. The weight cancels from the quotient and equality is attained. $\square$

Thus dividing both numerator and atomic diagonal by the number of shifts does not lower the exact support-only loss. At the first unbalanced TPC block $R = J$, the available ambient fiber bound from TPC-45 is

$$1 + \frac{L_0}{J} = X^{15077/105000 + o(1)}. \quad (129)$$

It exceeds $X^{1/400}$ by exponent

$$\frac{15077}{105000} - \frac{1}{400} = \frac{29629}{210000}. \quad (130)$$

The actual squared norm is the realized maximum fiber size, which can be smaller than this ambient upper bound. The theorem says only that shift weights alone do not reduce that realized norm.

9 Alias folding, local geometry, and fixed-shift localization

The frozen shift variable has a useful Fourier theory because the source–mask coefficient does not move with the intercept. The two shift families available from the inherited TPC construction have opposite defects: the completed aliases range over a macroscopic determinant window but are sampled on a very sparse progression, whereas genuinely reconstructed local-shift packets can presently be placed in one ambient geometry only on a much shorter interval. This section quantifies both statements and then records the sharp price of localizing a weighted average back to one prescribed shift.

9.1 The exact completed-alias fold

Let M_{al} and $\mathcal{I}_B$ be the inherited completion parameters, so that

$$M_{\text{al}} = X^{399/400 + o(1)}, \quad c_k = h_0 + kM_{\text{al}}, \quad K_B := |\mathcal{I}_B| = X^{1/400 + o(1)}. \quad (131)$$

For a determinant atom

$$x = (p, r, m, j)$$

write

$$\Delta(x) := pr - mj. \quad (132)$$

The following identity is the exact form of alias folding; it uses no estimate.

Lemma 9.1 (Weighted alias selector). *For arbitrary weights $(\omega_k)_{k \in \mathcal{I}_B}$ and any two integer determinant values Δ, Δ' ,*

$$\begin{aligned} \sum_{k \in \mathcal{I}_B} \omega_k \mathbf{1}_{\Delta = c_k} \mathbf{1}_{\Delta' = c_k} \\ = \mathbf{1}_{\Delta = \Delta'} \mathbf{1}_{\Delta \equiv h_0 \pmod{M_{\text{al}}}} \mathbf{1}_{(\Delta - h_0)/M_{\text{al}} \in \mathcal{I}_B} \omega_{(\Delta - h_0)/M_{\text{al}}}. \end{aligned} \quad (133)$$

Consequently, for every finitely supported coefficient field $A(x)$,

$$\begin{aligned} & \sum_{k \in \mathcal{I}_B} \omega_k \left| \sum_{\Delta(x)=c_k} A(x) \right|^2 \\ &= \sum_{\substack{x, x' \\ \Delta(x)=\Delta(x') \\ \Delta(x) \equiv h_0 \pmod{M_{\text{al}}} \\ (\Delta(x)-h_0)/M_{\text{al}} \in \mathcal{I}_B}} \omega_{(\Delta(x)-h_0)/M_{\text{al}}} A(x) \overline{A(x')}. \end{aligned} \quad (134)$$

Proof. At most one k can satisfy $\Delta = h_0 + kM_{\text{al}}$. It contributes to the left side of (133) precisely when the two determinant values coincide, lie in the required residue class, and their common quotient belongs to $\mathcal{I}_B$. Expanding the square proves (134). $\square$

Thus shift averaging does not turn the determinant form into its atomic diagonal. It replaces the fixed equation by an equal-determinant quartic correlation, restricted to one sparse residue class.

There is an equivalent frequency statement. For a finitely supported sequence $F : \mathbb{Z} \rightarrow \mathbb{C}$, define

$$\widehat{F}(\alpha) = \sum_{c \in \mathbb{Z}} F(c) e(-c\alpha), \quad \alpha \in \mathbb{T}, \quad (135)$$

and sample $G(k) = F(h_0 + kM_{\text{al}})$. Then, for every $\beta \in \mathbb{T}$,

$$\boxed{\widehat{G}(\beta) = \frac{1}{M_{\text{al}}} \sum_{a=0}^{M_{\text{al}}-1} e\left(\frac{h_0(\beta+a)}{M_{\text{al}}}\right) \widehat{F}\left(\frac{\beta+a}{M_{\text{al}}}\right)}. \quad (136)$$

Indeed, insertion of (135) into the right side gives

$$\sum_c F(c) e\left(\frac{(h_0 - c)\beta}{M_{\text{al}}}\right) \frac{1}{M_{\text{al}}} \sum_{a=0}^{M_{\text{al}}-1} e\left(\frac{(h_0 - c)a}{M_{\text{al}}}\right),$$

and the inner average selects $c = h_0 + kM_{\text{al}}$, leaving $\sum_k G(k) e(-k\beta)$. Multiplication by a finite alias window ω_k convolves the right side of (136) with $\widehat{\omega}$. It does not unfold the M_{al} frequency branches.

Remark 9.2 (What the inherited alias family represents). The common coefficient field in the alias family is exactly the one proved in TPC-42 and TPC-43 [5, 6]. Formula (136) is therefore an unconditional identity for the completed determinant packet. It is not the Fourier transform of a family of prime-pair problems reconstructed at shifts $h_0 + kM_{\text{al}}$. In particular, the row coprimality and fixed residue masks have not been recomputed at those integers.

9.2 Common embedding and a collision-free local range

We now vary the physical shift itself. Let

$$\mathfrak{H}_X \subset \mathbb{Z} \setminus \{0\}, \quad H_* := \max_{\nu \in \mathfrak{H}_X} |\nu - h_0|, \quad \widetilde{H}_* := |h_0| + H_*. \quad (137)$$

Choose ambient geometric row systems $\mathcal{A}_1^*, \mathcal{A}_2^*$ containing every row in the relevant dyadic source windows, without imposing shift-dependent coprimality. The actual packet at ν is embedded by setting a row to zero unless $(d_\rho, \nu) = 1$, and by setting each terminal atom to zero off its own support. The same zero extension is made on the opposite-row output coordinate. This places all finite packets in a common coordinate space; it does not make their coefficient values equal.

For $\rho = (\ell_\rho, d_\rho)$, define the local equality label

$$u_{\rho,j}^{(\nu)} := d_\rho(\ell_\rho d_\rho j + \nu). \quad (138)$$

Let D_+ be the largest ambient source divisor and J_- the smallest orbit time.

Lemma 9.3 (Uniform local-shift label rigidity). *Suppose*

$$\tilde{H}_* D_+ < J_-. \quad (139)$$

For every fixed $(\nu, j) \in \mathfrak{H}_X \times \mathcal{J}$, two active rows in the same source system satisfy

$$u_{\rho,j}^{(\nu)} = u_{\sigma,j}^{(\nu)} \implies \rho = \sigma. \quad (140)$$

At the TPC endpoint, every fixed $\epsilon > 0$ permits

$$H_* \leq X^{29629/210000-\epsilon}. \quad (141)$$

Proof. Writing $\rho = (\ell, d)$ and $\sigma = (\ell', e)$, equality of the labels gives

$$j(\ell d^2 - \ell' e^2) = \nu(e - d). \quad (142)$$

If the integer in parentheses is nonzero, the left side has magnitude at least $j \geq J_-$, whereas the right side has magnitude at most $\tilde{H}_* D_+ < J_-$. Hence both sides vanish. Since $\nu \neq 0$, first $d = e$, and then $\ell = \ell'$.

For the exponent, use

$$\frac{133}{400} = \frac{69825}{210000}, \quad \frac{10049}{52500} = \frac{40196}{210000}.$$

Their difference is $29629/210000$. Since h_0 is fixed, $\tilde{H}_* = H_* + O_{h_0}(1)$; the fixed ϵ absorbs this constant and the endpoint $o(1)$ -terms. $\square$

Since (141) is $o(X)$, common target endpoints remain $N_\pm^{\mathfrak{H}} = X^{1+o(1)}$ after harmless dyadic enlargement. A common large-prime band can therefore be defined from the minimum and maximum over the entire local family. The label rigidity lemma also shows that zero extension itself creates no same-coordinate collision. It does not control the dependence of the coefficient field on ν .

The support-only cofactor bound acquires an explicit dependence on the window. Suppose a touched target has $n = pr$, residual label $s = dr$, and cofactor $r \asymp R_0$. For fixed (ν, j, s) and $d \mid s$, the source rows solve

$$rp - dj\ell = \nu. \quad (143)$$

Putting $g = (r, dj)$, one has $g \mid \nu$. The source-prime step in the integral ladder is r/g , so a source window of length $O(L_0)$ contains

$$O\left(1 + \frac{gL_0}{r}\right) \leq O\left(1 + \frac{\tilde{H}_* L_0}{R_0}\right)$$

parameters. Summation over $d \mid s$ proves the uniform bound

$$\rho_{R_0, \mathfrak{H}} := \max_{\nu \in \mathfrak{H}_X, j \in \mathcal{J}, s} \# \left\{ (d, \ell, p, r) : \begin{array}{l} r \asymp R_0, \ dr = s, \\ rp - dj\ell = \nu, \text{ all packet supports hold} \end{array} \right\}$$

$$\boxed{\rho_{R_0, \mathfrak{H}} \ll \tau(s) \left(1 + \frac{\tilde{H}_* L_0}{R_0}\right)}. \quad (144)$$

At the balanced scale $R_0 \asymp L_0$, the presently available support-only bound therefore carries a polynomial factor for a polynomial local window. TPC-36 contains a stronger primitivity statement at an earlier ultra-divisor interface, but the terminal equality interface imported through TPC-43–TPC-45 does not state its preservation. We consequently retain (144); setting $g = 1$ would require a new provenance lemma.

There is a second uniformity issue. For each fixed ν , let $\mathfrak{M}_B(\nu; X)$ denote the least weighted projective mass in the analogue of (26). The inherited statement is only

$$\mathfrak{M}_B(\nu; X) \ll_{B, \nu} X^{o(1)} \quad \text{for every fixed } \nu. \quad (145)$$

It does not imply

$$\sup_{\nu \in \mathfrak{H}_X} \mathfrak{M}_B(\nu; X) = X^{o(1)} \quad (146)$$

when $\mathfrak{H}_X$ grows with X . The period $|\nu|$ of the fixed residue factor, the eligible rows, and the source residue data all move with ν . For a coherent weighted average, the corresponding sufficient condition is

$$\sum_{\nu \in \mathfrak{H}_X} |\omega_\nu| \mathfrak{M}_B(\nu; X) = X^{o(1)}.$$

Neither growing-family statement follows from TPC-36. A fixed finite set of physical shifts is harmless, but it supplies no mesoscopic shift average.

9.3 Why the proved collision-free range misses the Poisson scale

The frozen determinant family has a clean shift transform. For one scalar product-ready layer, put

$$F(c) = \sum_{p, r, m, j} a(p)b(r)c_0(m)w(j/J)\mathbf{1}_{pr-mj=c}. \quad (147)$$

Its full shift Fourier transform contains the exact orbit kernel

$$\begin{aligned} \sum_{j \in \mathbb{Z}} w(j/J) e(mj\alpha) &= J \sum_{k \in \mathbb{Z}} \widehat{w}(J(k - m\alpha)) \\ &=: K_m(\alpha). \end{aligned} \quad (148)$$

Suppose, for clarity, that $\widehat{w}$ is supported in a fixed interval. Since $m \asymp Q$ and $mJ \asymp X$, the zero Poisson packet has width $X^{-1+o(1)}$, while the first nonzero sampling centers lie at distance $Q^{-1+o(1)}$. Thus the geometric gap is

$$X^{-1+o(1)} < \|\alpha\|_{\mathbb{T}} < X^{-267/400+o(1)}. \quad (149)$$

To see the required shift length concretely, take a standard modulated weight

$$W_{H, \alpha_0}(c) = V\left(\frac{c - c_0}{H}\right) e(\alpha_0 c). \quad (150)$$

Its Fourier multiplier has width $\asymp H^{-1}$ around α_0 . Placing a fixed-shape multiplier inside the gap near its coarse Q^{-1} -end requires

$$H \gg Q = X^{267/400+o(1)}. \quad (151)$$

In contrast, the presently proved collision-free local-label range reaches only

$$H_* \leq X^{29629/210000-\epsilon}.$$

The exponent deficit to the first Poisson scale is

$$\frac{267}{400} - \frac{29629}{210000} = \frac{110546}{210000} = \frac{55273}{105000} > 0. \quad (152)$$

Hence the dense actual local family, under this rigidity argument, does not reach the scale on which the first sampling aliases can be spectrally resolved by this route. The completed-alias family spans a determinant interval of order X , but its spacing is $M_{\text{al}} = X^{399/400+o(1)}$; the exact fold (136) shows why macroscopic span does not replace dense sampling. The frozen completed family is the only one of the three current objects that simultaneously permits dense shifts on the Poisson scale and retains one coefficient field, and it is not a family of actual reconstructed TPC packets.

9.4 Weighted grouping and the sharp localization price

Shift averaging also does not improve the support norm merely by renormalizing numerator and diagonal with the same weights. Theorem 8.2 proves that the exact grouping norm remains the largest realized fiber: the common weight cancels on each block. An analytic averaged estimate may of course beat this support norm by using coefficient cancellation. A separate loss appears when that average is localized back to one prescribed shift.

Lemma 9.4 (Sharp weighted localization cost). *Let $E_\nu \geq 0$, let $\omega_\nu \geq 0$, and suppose $\omega_{\nu_0} > 0$. Define*

$$\text{Av}_\omega E := \frac{\sum_\nu \omega_\nu E_\nu}{\sum_\nu \omega_\nu}. \quad (153)$$

Then

$$E_{\nu_0} \leq \mathcal{L}_\omega(\nu_0) \text{Av}_\omega E, \quad \mathcal{L}_\omega(\nu_0) := \frac{\sum_\nu \omega_\nu}{\omega_{\nu_0}}. \quad (154)$$

The constant is sharp for arbitrary nonnegative data.

Proof. The term $\omega_{\nu_0} E_{\nu_0}$ is bounded by the full weighted sum, which proves the inequality. Equality holds when every E_ν except E_{ν_0} is zero. $\square$

For K essentially equal weights, $\mathcal{L}_\omega(\nu_0) \asymp K$, independently of whether the weights are normalized to have total mass one. In particular, a fixed-shape nonnegative smooth weight spread across K_B inherited aliases, with central value bounded below, has

$$\mathcal{L}_\omega(0) = X^{1/400+o(1)} \quad (155)$$

when its central value is bounded below. This is precisely the size of the TPC endpoint allowance K_B . Therefore an alias-averaged mean-square theorem at its averaged atomic scale does not, by positivity alone, yield the physical $k = 0$ bound at the same scale.

Remark 9.5 (Claim boundary). The identities above prove four different facts: exact folding of the completed aliases, a collision-free label range inside the common zero-extended geometry, a strict mismatch between that rigidity range and the Poisson resolution scale, and the sharp positivity cost of returning from an average to one shift. They do not show that localization is impossible. A maximal theorem, shift regularity, an amplifier, or a coefficient-specific comparison between neighboring shifts could reduce the loss. None is supplied by TPC-36, TPC-42, TPC-43, or TPC-45. Accordingly, no averaged or frozen-family result in this paper is asserted to close the physical fixed-shift TPC estimate, break the parity barrier, or imply the twin-prime conjecture.

10 The external analytic boundary

The deterministic reductions in this paper should not be conflated with an available Kloosterman–fraction theorem. The distinction is already visible at the scalar level and becomes sharper after the physical Hilbert output is restored. We record the coefficient slots, the role of shift averaging, and the normalization supplied by the relevant primary sources.

10.1 Three Kloosterman sequences are not three determinant slots

Let $\mathcal{A} = [A/2, A]$, $\mathcal{M} = [M/2, M]$, and $\mathcal{N} = [N/2, N]$. Bettin and Chandee’s trilinear form is

$$\mathcal{B}(M, N, A) := \sum_{\substack{a \in \mathcal{A}, m \in \mathcal{M}, n \in \mathcal{N} \\ (m, n) = 1}} \nu_a \alpha_m \beta_n e\left(\vartheta \frac{a\bar{m}}{n}\right), \quad \vartheta \neq 0, \quad (156)$$

where $\bar{m}$ is the inverse of m modulo n . Their Theorem 1 gives, for arbitrary complex sequences α, β, ν ,

$$|\mathcal{B}(M, N, A)| \ll \|\alpha\|_2 \|\beta\|_2 \|\nu\|_2 \left(1 + \frac{|\vartheta|A}{MN}\right)^{1/2} \times \left\{ (AMN)^{7/20+\varepsilon} (M+N)^{1/4} + (AMN)^{3/8+\varepsilon} (AN+AM)^{1/8} \right\}. \quad (157)$$

Thus (156) has three arbitrary sequences, but one of them is a sequence in the *Kloosterman numerator* a . This is not the same assertion as allowing three arbitrary weights on variables in a determinant equation [1, Theorem 1].

The distinction is explicit in the fixed-determinant consequence of the same paper. For

$$\mathcal{T}_\Delta := \sum_{\substack{m_i \in [M_i/2, M_i], n_i \in [N_i/2, N_i] \\ m_1 n_2 - m_2 n_1 = \Delta}} f(m_1) g(m_2) \alpha_{n_1} \beta_{n_2}, \quad \Delta \neq 0, \quad (158)$$

Corollary 1 assumes

$$f^{(j)} \ll \eta^j M_1^{-j}, \quad g^{(j)} \ll \eta^j M_2^{-j} \quad (j \geq 0), \quad (159)$$

and places arbitrary sequences only on n_1, n_2 . With

$$R_* = \frac{M_1 N_2}{M_2 N_1} + \frac{M_2 N_1}{M_1 N_2}, \quad (160)$$

the conclusion is

$$\begin{aligned} \mathcal{T}_\Delta = & \sum_{\substack{n_i \in [N_i/2, N_i] \\ (n_1, n_2) | \Delta}} \frac{(n_1, n_2)}{n_1 n_2} \alpha_{n_1} \beta_{n_2} \int_{\mathbb{R}} f\left(\frac{x+\Delta}{n_2}\right) g\left(\frac{x}{n_1}\right) dx \\ & + O\left((\eta R_*)^{3/2} \|\alpha\|_2 \|\beta\|_2 (N_1 N_2)^{7/20} (N_1 + N_2)^{1/4+\varepsilon} (M_1 M_2)^\varepsilon\right). \end{aligned} \quad (161)$$

Both smooth slots in (159) are part of the hypothesis; they are not cosmetic remnants of the proof [1, Corollary 1].

For the TPC equation

$$pr - mj = h, \quad (162)$$

one favorable identification is

$$(m_1, n_2, m_2, n_1) = (p, r, j, m). \quad (163)$$

It places the cofactor and source weights in the two arbitrary slots and the orbit weight in one smooth slot, but it places the prime weight in the other smooth slot. Interchanging p and r merely replaces the missing prime slot by a missing Möbius slot. Other identifications put the nonsmooth source coefficient in a smooth slot. Since one factor from pr and one factor from mj must occupy the two smooth positions, and only the envelope in j is intrinsically smooth, every identification loses at least one of the three arithmetic weights p, r, m . The actual fixed residue selector in j , retained in [equation \(32\)](#), only makes direct applicability more restrictive. This is the exact coefficient-slot obstruction isolated geometrically in TPC-45 [\[7\]](#).

10.2 What determinant-shift averaging actually buys

There is a genuine use of determinant-shift averaging in the work of Bettin, Chandee, and Radziwiłł. After their smooth decomposition and Poisson summation, a relevant off-diagonal block contains, schematically,

$$\sum_{n_1, n_2} \sum_{0 < |\Delta| \leq D/d} \sum_{0 < k < H_d} a_{dn_1} \overline{a_{dn_2}} e\left(-k\Delta \frac{\overline{n_1}}{n_2}\right) \mathcal{W}_{k, \Delta, n_1, n_2}, \quad (164)$$

where $\mathcal{W}$ is supplied by the already smooth variables. They set

$$b = k\Delta, \quad \nu_{x,y}(b) := \sum_{k\Delta=b} e\left(-kx + \frac{\Delta y}{2\pi}\right). \quad (165)$$

Consequently

$$|\nu_{x,y}(b)| \leq \tau(|b|), \quad \|\nu_{x,y}\|_2 \ll B^{1/2+\varepsilon} \quad (|b| \ll B), \quad (166)$$

and the pair (k, Δ) becomes the single numerator sequence in [\(156\)](#). Bettin–Chandee cancellation is then available in that numerator range [\[2, Section 5\]](#), especially the reduction preceding their small-frequency estimate].

This mechanism says exactly what the average provides: it lengthens and structures the Kloosterman-numerator sequence. It is available only after Poisson summation has used smooth determinant variables. It does not replace either smooth hypothesis in [\(159\)](#), and it does not turn [\(161\)](#) into a determinant theorem with arbitrary weights on p, r, m . An arbitrary external weight on Δ can also increase the multiplicative energy of the compressed sequence in [\(165\)](#); its ℓ^2 norm must be proved rather than silently identified with the product of two norms.

10.3 Nearby dispersion estimates

Fouvry and Radziwiłł study a different average. For divisor-bounded sequences α_m, β_n , they average the congruence discrepancy

$$\sum_{q \asymp Q} \left| \sum_{mn \equiv a \pmod{q}} \alpha_m \beta_n - \frac{1}{\varphi(q)} \sum_{(mn, q)=1} \alpha_m \beta_n \right| \quad (167)$$

over the modulus q , while the residue $a \neq 0$ is fixed. Their first range, under the stated Siegel–Walfisz hypothesis on the short sequence, is

$$\exp((\log X)^\varepsilon) \leq N \leq Q^{-11/12} X^{17/36-\varepsilon}. \quad (168)$$

Their prime-modulus variant removes that Siegel–Walfisz input in the same power range [\[3, Corollary 1.1 and Theorem 1.2\]](#). Wright’s partially fixed-modulus theorem estimates

$$\sum_{\substack{a \asymp A, m \asymp M, n \asymp N \\ (m, nR)=1}} \nu_a \alpha_m \beta_n e\left(\vartheta \frac{a\overline{m}}{nR}\right) \quad (169)$$

and improves the resulting unbalanced-convolution ranges, including $N \leq Q^{-33/28} X^{17/28-\varepsilon}$ in the first regime [8, Theorem 2.1 and Corollary 2.2]. These are genuine advances for modulus dispersion and fixed denominator factors. Neither theorem averages the physical determinant shift in (162), supplies the missing determinant coefficient slot, or controls the inherited TPC masks after successive physical restrictions.

10.4 Why a scalar black box would still not prove the atomic estimate

The external results above are scalar. The TPC target inherited from the projective and residual-fiber reductions is an atomic-energy estimate for a Hilbert-valued synthesis, not merely a scalar count [4–7]. Formally one may write

$$\left\| \sum_e x_e \right\|_{\mathcal{H}} = \sup_{\|v\|_{\mathcal{H}}=1} \left| \sum_e \langle x_e, v \rangle_{\mathcal{H}} \right|. \quad (170)$$

But the scalar coefficient $\langle x_{p,r,m,j}, v \rangle$ produced by (170) generally couples all four indices. It need not factor into the separate sequences required by (156) or (158). Uniformity of (157) over each of its three sequences does not imply uniformity over such an entangled four-index tensor.

There is a second normalization mismatch. Squaring a scalar estimate and summing over ambient coordinates naturally returns a Cartesian tensor energy, or after shift averaging the sum of diagonals over all shifts. The physical TPC denominator retains only the atomic diagonal at the prescribed shift and after the actual projective identifications. TPC–36 and TPC–42 show why these identifications and coherent defects cannot be discarded; TPC–43 and TPC–45 show why physical sign restriction and large-prime descent must be applied to one globally consistent coefficient packet. None of (157), (161), (167), or (169) contains this atomic normalization or proves stability under those restrictions.

We therefore use the cited literature only for comparison. It does not prove the vector atomic estimate studied here. Conversely, the slot and normalization mismatches are not impossibility theorems: a future positive result could exploit the special prime/Möbius packet, prove a restriction-stable tensor estimate, or establish a separate localization principle. Such an input would be new and would have to state these features explicitly.

11 The next gate

The preceding theorems split the route beyond TPC–45 into four logically different tasks.

11.1 Orbit-frequency regularity of the actual coefficient

The spectral gap in section 6 is stable under a Hilbert vector $v_{p,r,m}$ that is independent of j , or more generally under a uniformly band-limited j -dependence. The actual dual coefficient has the form

$$z_{\text{output row}, j, d(m)r}. \quad (171)$$

No theorem above bounds its orbit bandwidth uniformly over the unit ball of the dual output space. A positive continuation would therefore need either

- (i) a restriction-stable frequency decomposition of the actual mask and output field, with projective mass $X^{o(1)}$; or
- (ii) a vector-valued sampling theorem that controls (171) without first replacing it by a scalar smooth profile.

The fixed-column decomposition of TPC–36 supplies the scalar smooth profile before dualization, not either of these stronger conclusions.

11.2 Atomic, rather than Cartesian, normalization

The fixed-shift tensor theorem naturally gives

$$X^{o(1)}\mathcal{D}_{\otimes} = X^{o(1)}\sum_h \Delta_h. \quad (172)$$

The physical target instead asks for a bound on the synthesis at one shift by

$$X^{1/400+o(1)}\Delta_{h_0} \quad (173)$$

after the actual output identifications. The examples in [section 8](#) show that no theorem for arbitrary three coefficient slots can bridge (172) and (173). The next arithmetic input must use the actual source phases, the projective masks, or cancellation between residual fibers. Merely retaining Möbius symbols is insufficient: TPC-45 proves that their product is constant on each equal-residual fiber.

11.3 Alias unfolding and localization

The TPC alias family samples the determinant at spacing $M_{\text{al}} = X^{399/400+o(1)}$. Its spectrum is the folded sum in [equation \(136\)](#); a gap in the completed consecutive-shift family does not survive that folding without an additional anti-alias estimate. Even an averaged estimate on a consecutive window does not select h_0 without a maximal or regularity theorem, and the localization factor in [lemma 9.4](#) is sharp.

A useful positive target is therefore a coefficient-specific inequality of the following schematic form, for $H_* \geq 1$:

$$\sup_{|h-h_0| \leq H_*} \frac{\|F_h\|^2}{\Delta_h + \delta_X} \ll X^{o(1)} \left[\frac{1}{H_*} \sum_{|h-h_0| \leq H_*} \frac{\|F_h\|^2}{\Delta_h + \delta_X} \right], \quad (174)$$

where fixed normalization constants in the window average are inessential. Here $\delta_X > 0$ is a declared regularizer; one also needs uniform packet provenance and a range H_* that overlaps the analytic averaging scale. Formula (174) is a target, not an assumption or a consequence of the present paper.

11.4 A reversible continuation

The next paper may pursue any of the following doors.

- Door A.** Prove orbit-frequency decay for one actual projected source/mask sector and transfer a nonempty part of the shift-spectral gap to that sector.
- Door B.** Establish an atomic-normalized estimate on one additional cofactor block, even after averaging a fixed finite set of output rows.
- Door C.** Prove an anti-alias or maximal-in-shift theorem connecting the completed family to the sparse TPC alias family.
- Door D.** Construct actual-mask coefficients that saturate the tensor-to-atomic or fiber frontier. This would give a sharp negative boundary for the present route without asserting that the full physical coefficient is large.

Success at any door is a genuine advance. Failure at one does not invalidate the spectral gap, tensor closure, or sharp boundary results proved here. None of these doors may be relabelled as a fixed-shift prime-pair asymptotic or a breach of the parity barrier.

12 Exact certificate and claim ledger

The reproducibility directory contains

`experiments/tpc46_certificate.py`.

It uses only the Python standard library. Integer and Gaussian-integer arithmetic is exact; rational exponent comparisons use `fractions.Fraction`; and all pseudo-random trials use the fixed seed 4601. A successful run writes `experiments/tpc46_certificate.json`.

The current certificate performs 122232 exact checks. Its canonical payload hash is

`38287f0cbee8060ebfb73407f6a4cd0255490e64effefea4481ee67ec1adb4ea`.

The hash covers the canonical JSON report before the hash field is appended. The certificate checks:

- (i) exact equality between the four-variable determinant synthesis and the additive correlation of the two product sequences, on 640 deterministic Gaussian-integer trials;
- (ii) exact disintegration of Cartesian tensor energy into the atomic diagonals over all shifts, together with the multiplicity-based scalar tensor bound;
- (iii) injectivity of the large-prime product map on a finite supported model;
- (iv) the residual collision identity and a finite Walsh scalarization of the squarefree physical labels $s = d(m)r$;
- (v) the exact weighted grouping norm and its maximizing fiber;
- (vi) the prescribed-shift coherent prime construction;
- (vii) exact finite-field DFT analogues of the orbit spectral-support theorem and alias-frequency folding, together with the sharp localization spike; and
- (viii) the principal rational endpoint, budget, coordinate, and scale differences listed in the generated report.

The finite Walsh check uses squarefree residual labels, as at the nonzero physical equality atoms; the analytic Bohr-torus theorem for general integer labels is proved separately in the text. The finite cyclic spectral checks are algebraic finite-field DFT analogues; they do not certify the continuous Poisson theorem. Likewise, finite large-prime injectivity does not certify an asymptotic prime-counting statement. Those facts are proved in the text from their explicit hypotheses. The certificate does not test Möbius cancellation, Fourier decay of the actual dual output, a growing-shift projective decomposition, an anti-alias estimate, or a prime-pair asymptotic.

For clarity, the logical status of the main conclusions is as follows.

1. The physical large-prime determinant expansion and the exact pulled-back dual field in [section 2](#) are conditional only on the cited TPC coefficient packet and its declared dyadic support. No new cancellation hypothesis is inserted.
2. The scalar tensor closure, residual Fourier–Bohr closure, shift-kernel factorization, spectral gap, Poisson identities, grouping norm, alias fold, and localization inequality are unconditional theorems for the finite or smooth models stated in their hypotheses.

3. The band-limited exact gap applies to a frozen packet whose Hilbert coefficient is independent of j , or to a separately declared band-limited extension. A fixed residue-class restriction preserves the endpoint exponents under the hypotheses of [corollary 6.3](#). For ordinary compactly supported smooth profiles, the conclusion is rapid decay on a shrunk band, not exact vanishing.
4. The tensor bound is normalized by $\mathcal{D}_{\otimes} = \sum_h \Delta_h$. It is not a bound at the physical atomic scale Δ_{h_0} . The counterexamples concern arbitrary separated slots and do not assert that the actual TPC source/mask realizes their coherent phases.
5. The completed alias bank is a common-coefficient auxiliary family, not a collection of reconstructed prime-pair problems at the sampled integers. The local physical family always has a zero-extended common ambient coordinate space; only its collision-free label range is restricted by the displayed rigidity argument. The inherited projective theorem is not asserted uniformly for a growing set of shifts.
6. The external literature audit shows that the cited formulations do not directly supply the current coefficient and normalization interface. This is not an impossibility theorem for every future analytic method.
7. The doors in [section 11](#) are proposed next inputs. None is assumed or proved here.
8. No statement implies fixed-shift TPC closure, Chowla at a fixed shift, the Hardy–Littlewood prime-pair conjecture, a breach of the parity barrier, or the twin-prime conjecture.

The positive advance is therefore exact but model-specific: the smooth orbit slot creates a real mesoscopic gap in the completed shift spectrum, while the tensor and Bohr calculations identify precisely which energy they control. The remaining physical transfer gaps are displayed rather than promoted to hidden hypotheses.

References

- [1] Sandro Bettin and Vorrapan Chandee. Trilinear forms with Kloosterman fractions. *Advances in Mathematics*, 328:1234–1262, 2018. doi: 10.1016/j.aim.2018.01.026. URL <https://doi.org/10.1016/j.aim.2018.01.026>.
- [2] Sandro Bettin, Vorrapan Chandee, and Maksym Radziwiłł. The mean square of the product of the Riemann zeta-function with Dirichlet polynomials. *Journal für die reine und angewandte Mathematik*, 729:51–79, 2017. doi: 10.1515/crelle-2014-0133. URL <https://doi.org/10.1515/crelle-2014-0133>.
- [3] Étienne Fouvry and Maksym Radziwiłł. Level of distribution of unbalanced convolutions. *Annales Scientifiques de l’École Normale Supérieure*, 55(2):537–568, 2022. doi: 10.24033/asens.2501. URL <https://doi.org/10.24033/asens.2501>.
- [4] Liang Wang. Soft projective reassembly of the physical row mask: Orthogonal GCD compression, optimal Ferrers separation, and coherent product-slope spectra. Preprint, repository commit `ecbd29afd77b`, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/ecbd29afd77b92224aa7720c11c65444407fc0e9/papers/tpc-36-soft-projective-mask-reassembly.
- [5] Liang Wang. Residue-fiber orthogonality at the triple-prime Möbius gate: Canonical Hilbert defects, coherent synthesis, and exact convolution counterterms. Preprint, repository commit `b99951f887ae`, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/b99951f887aecc977d0bb2af2431e32109332d05/papers/tpc-42-residue-fiber-coherent-defect.
- [6] Liang Wang. Annealed multiplicative closure at the four-Möbius gate: Prime-sign orthogonality, product-fiber rigidity, and the deterministic Walsh corner. Preprint, repository commit `301f7fc40cd1`, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/301f7fc40cd1.

301f7fc40cd1f08447feaa323766493b323b4106/papers/
tpc-43-annealed-multiplicative-closure.

- [7] Liang Wang. Large-prime cofactor descent for a four-Möbius Walsh energy. Preprint, repository commit 3034ff394aa3, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/3034ff394aa376ac08a64561c0306d7eff2f3579/papers/tpc-45-large-prime-cofactor-descent.
- [8] Thomas Wright. Trilinear Kloosterman fractions I: Partially fixed moduli and unbalanced convolutions. Preprint, 2026. URL <https://arxiv.org/abs/2604.25177>.